INTEGRALS

The Fundamental Theorem of Calculus

Differentiation undoes integration; the computational engine of integral calculus.

Part 1

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

- f continuous
- Every continuous function has an antiderivative: its accumulation function $F(x) = \int_a^x f(t)\,dt$

GIVES AN ANTIDERIVATIVE →

Part 2

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

- F any antiderivative of f
- Written $F(x)\Big|_a^b$ or $[F(x)]_a^b$

EVALUATE AT THE ENDPOINTS →

Example

$$\int_1^3 x^2\,dx = \frac{x^3}{3}\Big|_1^3 = \frac{27}{3} - \frac{1}{3} = \frac{26}{3}$$

- Upper limit first: $F(3) - F(1)$
- The constant of integration cancels
- $x^2 > 0$ on $[1, 3]$, so the result is positive